Seminar 2 (8.10.2025)

Conversions

b = source base
h = destination base

Substitution method	Successive divisions & multiplications
→ computations in destination base (h)	→ computations in source base
→ convert the digits in source representation in base h	→ int. part: divisions by h
→ convert the base b to base h	→ fr. part: multiplications by h
$\boxed{b < h}$	$\boxed{b > h}$
→ compute the sum.	

① $2321,23_{(4)} = ?_{(7)}$ (Substitution $4<7$)

$$2321,23_{(4)} = 1_{(7)} \cdot 4^0_{(7)} + 2_{(7)} \cdot 4^1_{(7)} + 3_{(7)} \cdot 4^2_{(7)} + 2_{(7)} \cdot 4^3_{(7)} +$$
$$+ 2_{(7)} \cdot 4^{-1}_{(7)} + 3_{(7)} \cdot 4^{-2}_{(7)}.$$

$1_{(7)} \cdot 4^0_{(7)} = 1_{(7)}$

$2_{(7)} \cdot 4_{(7)} = 11_{(7)}$

$2 \cdot 4 = 8 \,/\, 7 = 1$
$\%7 = 1$

$3_{(7)} \cdot 4^2_{(7)} = 3_{(7)} \cdot 22_{(7)} = 66_{(7)}$

$$\begin{array}{r} 4_{(7)} \cdot \\ 4_{(7)} \\ \hline 22_{(7)} \end{array} \qquad 4 \cdot 4 = 16 \,/\, 7 = 2,\ \%7 = 2 \qquad \begin{array}{r} 22_{(7)} \\ 3_{(7)} \\ \hline 66_{(7)} \end{array}$$

$2_{(7)} \cdot 4^3_{(7)} = 242_{(7)}$

$$\begin{array}{r} \overset{1}{2}2_{(7)} \cdot \\ 4_{(7)} \\ \hline 121_{(7)} \end{array} \qquad 1 + 2 \cdot 4 = 9 \,/\, 7 = 1,\ \%7 = 2 \qquad \begin{array}{r} 121_{(7)} \cdot \\ 2_{(7)} \\ \hline 242_{(7)} \end{array}$$

$2_{(7)} \cdot 4^{-1}_{(7)} = 0,33$

$$2,00_{(7)} \ \big|\ 4_{(7)} \qquad \qquad 20_{(7)} = 14 \,/\, 4 = 3$$
$$\qquad \quad 0,333\ldots \qquad \qquad \%4 = 2$$

$$\begin{array}{r} 20 \\ 1 \\ \hline 20 \end{array}$$

$3_{(7)} \cdot 4^{-2}{}_{(7)} = 0,12$

$$\begin{array}{l|l} 3,00_{(7)} & 4_{(7)} \\ \hline 1 & 0,515151\ldots \\ \hline 30 & \\ 1 & \\ \hline 10 & \\ 1 & \\ \hline 3 & \end{array}$$

$30_{(7)} = 21 / 4 = 5$
$\% 4 = 1$

$10_{(7)} = 7 / 4 = 1$
$\% 4 = 3$

$$\begin{array}{l|l} 0,51_{(7)} & 4_{(7)} \\ \hline 1 & 0,12_{(7)} \\ \hline 5 & \\ 1 & \\ \hline 11 & \\ 1 & \\ \hline 0 & \end{array}$$

$11_{(7)} = 1+7 = 8 / 4 = 2$
$\% 4 = 0$

$2321_{(4)} = 1_{(7)} + 11_{(7)} + 66_{(7)} + 242_{(7)} = 353_{(7)}$

$$\begin{array}{r} 242_{(7)} \\ 66_{(7)} \\ 11_{(7)} \\ 1_{(7)} \\ \hline 353_{(7)} \end{array} \ (+)$$

$2+6+1+1 = 10 / 7 = 1$
$\% 7 = 3$

$1+4+6+1 = 12 / 7 = 1$
$\% 7 = 5$

$0,23_{(4)} = 0,33_{(7)} + 0,12_{(7)} = 0,45_{(7)}$

$$\begin{array}{r} 0,33_{(7)} \\ 0,12_{(7)} \\ \hline 0,45_{(7)} \end{array} \ +$$

$\Rightarrow \boxed{2321,23_{(4)} = \underbrace{353}, \underbrace{45}_{(7)}}$

② $3251,53_{(6)} = ?_{(16)}$ (Substitution $6 < 16$)

$= 1_{(16)} \cdot 6^{0}{}_{(16)} + 5_{(16)} \cdot 6^{1}{}_{(16)} + 2_{(16)} \cdot 6^{2}{}_{(16)} + 3_{(16)} \cdot 6^{3}{}_{(16)} +$

$+ 5_{(16)} \cdot 6^{-1}{}_{(16)} + 3_{(16)} \cdot 6^{-2}{}_{(16)}$

$1_{(16)} \cdot 6^{0}{}_{(16)} = 1_{(16)}$ $\qquad$ $2_{(16)} \cdot 6^{2}{}_{(16)} = 2_{(16)} \cdot 24_{(16)} = 48_{(16)}$

$5(16) \cdot 6(16) = 1E(16)$

$30/16 = 1$

$\%16 = 14 = E$

$$\begin{array}{r} 5(16) \cdot \\ 6(16) \\ \hline 1E(16) \end{array}$$

$6(16) \cdot$
$6(16)$
$\overline{24(16)}$

$36/16 = 2$

$\%16 = 4$

$3(16) \cdot 6^3(16) = 3(16) \cdot D8(16) = 288(16)$

$$\begin{array}{r} 24(16) \cdot \\ 6(16) \\ \hline D8(16) \end{array}$$

$24/16 = 1$

$\%16 = 8$

$1+12 = 13 = D$

$$\begin{array}{r} D8(16) \cdot \\ 3(16) \\ \hline 288(16) \end{array}$$

$24/16 = 1$

$\%16 = 8$

$1 + 13 \cdot 3 = 40$

$40/16 = 2$

$\%16 = 8$

$5(16) \cdot 6^{-1}(16) = 0{,}05$

$5{,}00(16) \mid 6(16)$

$0{,}D5$

1
50
1
20
1
2

$50(16) = 5 \cdot 16 = 80/6 = 13 = D$

$\%6 = 2$

$20(16) = 2 \cdot 16 = 32/6 = 5$

$\%6 = 2$

$3(16) \cdot 6^{-2}(16) = 0{,}15$

$3{,}00(16) \mid 6(16)$

$0{,}8(16)$

1
30
1
0

$30(16) = 3 \cdot 16 = 48/6 = 8$

$\%6 = 0$

$\downarrow$

$0{,}80(16) \mid 6(16)$

$0{,}15$

1
8
1
20
1
2

$20(16) = 32/6 = 5$

$\%6 = 2$

$3251(6) = 1(16) + 1E(16) + 48(16) + 288(16) = 2EF(16)$

1
$288 +$
48
$1E$

$8+8+E+1 = 17+14 = 31/16 = 1$

$\%16 = F$

$$\frac{1}{2EF_{(16)}}$$

$1+8+4+1 = 14 = E$

$$\boxed{0{,}53_{(6)} = 0{,}D5_{(16)} + 0{,}15_{(16)} = 0{,}EA_{(16)}}$$

$$\begin{array}{r} 0{,}D5_{(16)} \\ 0{,}15_{(16)} \\ \hline 0{,}EA_{(16)} \end{array} \; +$$

$$\Rightarrow \boxed{3251{,}53_{(6)} = 2EF{,}EA_{(16)}}$$

(3) $\underbrace{7527}_{\downarrow},\underbrace{65}_{\downarrow}{}_{(8)} = ?_{(5)}$ (Succesive $8>5$)

divide w destination base

multiply w destination base

$7527_{(8)} : 5_{(8)} = 1421_{(8)}$; 25, 12, 7, remainder 2

$1421_{(8)} : 5_{(8)} = 235_{(8)}$; 22, 31, remainder 0

$235_{(8)} : 5_{(8)} = 37_{(8)}$; 45, remainder 2

$37_{(8)} : 5_{(8)} = 6_{(8)}$; remainder 1

$6_{(8)} : 5_{(8)} = 1_{(8)}$; remainder 1

$1_{(8)} : 5_{(8)} = 0$; remainder 1

$25_{(8)} = 5+16 = 21 \;/5 = 4$
$\%5 = 1$

$12_{(8)} = 2+8 = 10 \;/5 = 2$
$\%5 = 0$

$7/5 = 1$
$\%5 = 2$

$14_{(8)} = 4+8 = 12 \;/5 = 2$
$\%5 = 2$

$22_{(8)} = 2+16 = 18 \;/5 = 3$
$\%5 = 3$

$31_{(8)} = 1+24 = 25 \;/5 = 5$
$\%5 = 0$

$23_{(8)} = 3+16 = 19 \;/5 = 3$
$\%5 = 4$

$45_{(8)} = 5+32 = 37 \;/5 = 7$
$\%5 = 2$

$37_{(8)} = 7+24 = 31 \;/5 = 6$
$\%5 = 1$

$0{,}65_{(8)} \cdot 5_{(8)} = 4{,}11_{(8)}$

$0{,}11_{(8)} \cdot 5_{(8)} = 0{,}55_{(8)}$

$0{,}55_{(8)} \cdot 5_{(8)} = 3{,}41_{(8)}$

$$\begin{array}{r} {}^{4\,3} \\ 0{,}65 \cdot \\ 5 \\ \hline 4{,}11_{(8)} \end{array}$$

$5 \cdot 5 = 25 \;/8 = 3$
$\%8 = 1$

$3+30 = 33 \;/8 = 4$
$\%8 = 1$

$$\Rightarrow \boxed{7527{,}65_{(8)} = \underline{111202},\underline{403}_{(5)}}$$

When do we stop multiplying? 3 cases
- given no of digits
- periodicity
- …,0

(4) D2A1,F6(16) = ? (7)

```
D2A1(16) | 7(16)
 1       |------
 --      | 1E17(16) | 7(16)
 62      |  1       |------
 1       |  --      | 44C(16) | 7(16)
 --      |  21      |  1      |------
  A      |  1       |  --     | 9D(16) | 7(16)
  1      |  --      |  5C     |  1     |------
  --     |  57      |  1      |  --    | 16(16) | 7(16)
  31     |  1       |  --     |  2D    |  1     |------
  1      |  --      |  1      |  1     |  --    |  3
  --     |  3       |         |  --    |  1
  0      |          |         |  3     |
```

62 (16) = 98 / 7 = 14 = E
%7 = 0.

1E (16) = E+16 = 14+16 = 30 / 7 = 4
%7 = 2

21 (16) = 1+16·2 = 33 / 7 = 4
%7 = 5

31 (16) = 49 / 7 = 7
%7 = 0

57(16) = 87 / 7 = 12 = C
%7 = 3

44 (16) = 4 + 4.16 = 68 / 7 = 9
%7 = 5

5C(16) = C+5·16 = 92 / 7 = 13 = D
%7 = 1

D2A1 = 313130

0,F6 (16) · 7(16) = 6,BA (16)

```
  5  3
  0,F6 (16) ·
      7 (16)
 ------------
  6,BA (16)
```

7·6 = 42 / 16 = 2
%16 = 10 = A

3 + 15·7 = 107 / 16 = 6
%16 = 11 = B

```
  5        2          3
  16 ·     16 ·       16 ·
   6        4          5
 ----     ----       ----
  96       64         80
```

0,BA (16) · 7 (16) = 5,16 (16)

```
  5  4
  0,BA (16) ·
      7 (16)
 ------------
  5,16 (16)
```

A·7 = 70 / 16 = 4
%16 = 6

4 + 77 = 81 / 16 = 5
%16 = 1

0,16 (16) · 7 (16) = 0,9A (16)

```
    2
  0,16 (16) ·
      7 (16)
 ------------
  0,9A (16)
```

6·7 = 42 / 16 = 2
%16 = 10 = A

2 + 7 = 9

$0,9A_{(16)} \cdot 7_{(16)} = 3,46_{(16)}$

$$\begin{array}{r} {}^{3\ 4} \\ 0,9A_{(16)} \\ 7_{(16)} \\ \hline 3,46_{(16)} \end{array}$$

$70 / 16 = 4$
$\% 16 = 6$

$4 + 63 = 67 / 16 = 4$
$\% 16 = 3$

$0,F6 = 0,6503\ldots ?$

$\Rightarrow \boxed{D2A1,F6_{(16)} = 313130,6503\ldots ?}$

(5) $CDA,23_{(16)} = ?_{(2)}$

$$\begin{array}{cc} \downarrow & \downarrow \\ 2^4 & 2^1 \end{array}$$

$2 \to 2^4 \Rightarrow$ groups of 4:

$CDA,23_{(16)} = \underbrace{1100}\ \underbrace{1101}\ \underbrace{1010},\ \underbrace{0010}\ \underbrace{0011}_{(2)}$

$2 \to 2^2 \Rightarrow$ groups of 2:

$CDA,23_{(16)} = \underbrace{11}\underbrace{00}\ \underbrace{11}\underbrace{01}\ \underbrace{10}\underbrace{10},\ \underbrace{00}\underbrace{10}\ \underbrace{00}\underbrace{11}_{(2)}$

$= 3\ 0\ \ 3\ 1\ \ 2\ 2,\ 0\ 2\ 0\ 3$

$= 303122,0203_{(4=2^2)}$

$2 \to 2^3 \Rightarrow$ groups of 3:

$CDA,23_{(16)} = \underbrace{110}\underbrace{011}\underbrace{010}\underbrace{11010},\ \underbrace{001}\underbrace{000}\underbrace{110}_{(2)}$

$= 6\ 3\ 3\ 2,\ 1\ 0\ 6$

$= 6332,106_{(8=2^3)}$

! Rapid conversions:

$2 \to 2^k$

$2^k \to 2$

$2 \to 16$
$\to 8$
$\to 4$
$\leftarrow$

! Do not use B10 as an intermediate base! Use B2.